Introduction

The research question for this final project is: which factors are most strongly associated with box office revenue for internationally released films from 2010 to 2024, and how do these relationships vary across genres? More specifically, I focus on production scale, audience engagement, genre, and franchise branding as the main candidate predictors of commercial success.

The outcome of interest is box office revenue. The central modeling question is whether revenue can be explained well with a mostly linear structure, or whether important nonlinear and interaction like patterns emerge once larger and more flexible models are introduced. This extends my midterm project, where budget, vote count, and genre already appeared to matter, but where the relationships were not fully stable across subgroups and the dataset was smaller.

The data come from The Movie Database (TMDB) API. For the final project, I compare three modeling approaches: a multiple linear regression model as an interpretable baseline, a generalized additive model to allow smooth nonlinear effects for continuous predictors, and XGBoost as a more flexible machine learning benchmark. Together, these models make it possible to compare interpretability, nonlinear structure, and predictive performance within the same cleaned dataset.

Methods

Movie IDs were collected from TMDB's discover/movie endpoint using a fixed release-date range and vote-count threshold, and then enriched using the /movie/{movie_id} endpoint to retrieve movie-level details such as budget, revenue, popularity, vote average, vote count, release date, genre metadata, and collection membership. The final cleaned dataset contains 4,462 movies, which is larger than the midterm sample and provides stronger support for genre level comparisons.

Several preprocessing decisions were necessary before modeling. First, I converted dates and numeric fields into appropriate data types and retained only released, non-adult films with non-missing and positive budget and revenue values. This matters because TMDB frequently records unavailable financial values as zero rather than true zero. Second, I applied log10 transformations to revenue and budget, and a log transformation to vote count and popularity, because these variables are strongly right-skewed and dominated by blockbuster-scale outliers in raw form. Third, I centered release year so that the model would be better scaled and easier to interpret. Finally, I collapsed infrequent genres into an Other category to avoid unstable coefficient estimates from very small groups.

The final feature set used for modeling includes log_budget, log_vote_count, log_popularity, vote_average, year_centered, primary_genre, and in_collection. The variable in_collection is a franchise proxy derived from TMDB collection membership and is intended to capture whether a film belongs to an established branded series. I used an 80/20 train-test split and compared models using out-of-sample RMSE, MAE, and R^2.

The first model was a multiple linear regression on log_revenue, fitted with HC3 robust standard errors to account for heteroskedasticity. Given that the features and target are in tabular form, it is a good starting point.

For the second model, I fit a generalized additive model to test whether the relationship between box office revenue and the main continuous predictors was more flexible than a standard linear model could capture. The motivation for this choice is because variables such as budget, vote count, popularity, and release year are unlikely to affect revenue in perfectly linear ways across their full range. By allowing smooth spline terms for these predictors, the GAM could capture diminishing returns or threshold-like behavior while remaining substantially more interpretable than a black-box model.

The final model I chose to explore XGBoost because it provides a strong overall predictive performance while still remaining practical for structured tabular data. Box office revenue is likely driven by nonlinear effects and interactions among budget, audience engagement, genre, and franchise status, and XGBoost is well suited to learning these relationships without requiring them to be specified manually in advance. Compared to black box neural net models, tree based models offer interpretability via its feature split significance. Compared with Random Forest, XGBoost typically produces stronger predictive performance because boosting builds trees sequentially to correct earlier errors rather than averaging many independent trees.

To tune my XGB model, I performed a grid search over boosting rounds, learning rate, tree depth, and subsampling parameters. Each candidate configuration was evaluated using 5 fold cross validation, and the final model was chosen based on the best average cross-validated RMSE on the training data set.

Results

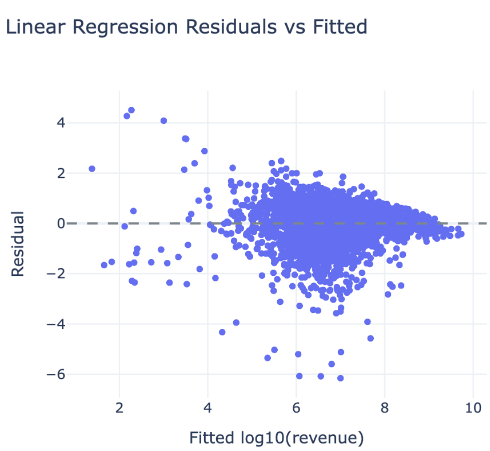
Across all models, production budget remained the single strongest predictor of box office revenue. This pattern was stable in the linear regression, the GAM, and XGBoost, which suggests that the budget-revenue relationship is the structural core of the dataset. Audience engagement, especially vote count, was the second strongest driver, while franchise membership (in_collection) also contributed meaningfully to predicted revenue levels.

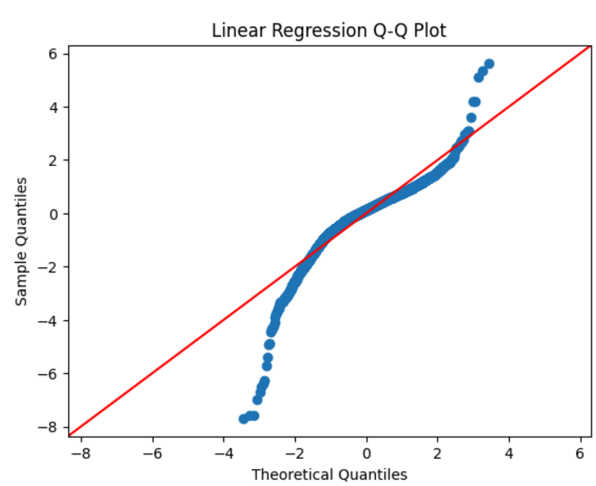
Linear Regression

I began my investigation with a linear regression model.Linear Regression OLS Summary

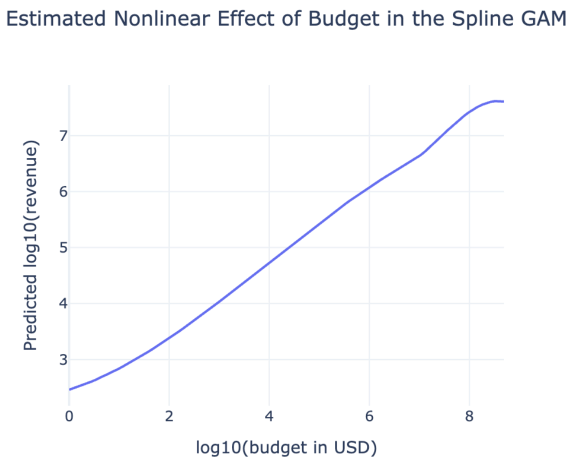
term		coef	std_err	stat	p_value	ci_low	ci_high

0	Intercept	0.10	0.30	0.33	0.742	-0.48	0.68
1	C(primary_genre)[T.Adventure]	0.02	0.06	0.28	0.782	-0.10	0.13
2	C(primary_genre)[T.Animation]	-0.02	0.06	-0.34	0.735	-0.14	0.10
3	C(primary_genre)[T.Comedy]	0.23	0.04	5.07	<0.001	0.14	0.31
4	C(primary_genre)[T.Crime]	-0.22	0.09	-2.43	0.015	-0.39	-0.04
5	C(primary_genre)[T.Drama]	-0.18	0.05	-3.94	<0.001	-0.28	-0.09
6	C(primary_genre)[T.Family]	0.08	0.10	0.85	0.395	-0.10	0.27
7	C(primary_genre)[T.Fantasy]	-0.09	0.10	-0.93	0.350	-0.29	0.10
8	C(primary_genre)[T.History]	-0.25	0.16	-1.55	0.121	-0.56	0.07
9	C(primary_genre)[T.Horror]	0.11	0.06	1.66	0.098	-0.02	0.23
10	C(primary_genre)[T.Other]	-0.07	0.09	-0.72	0.471	-0.24	0.11
11	C(primary_genre)[T.Romance]	0.18	0.08	2.25	0.024	0.02	0.33
12	C(primary_genre)[T.Science Fiction]	-0.09	0.08	-1.12	0.265	-0.23	0.06
13	C(primary_genre)[T.Thriller]	-0.20	0.08	-2.60	0.009	-0.35	-0.05
14	C(primary_genre)[T.War]	-0.20	0.16	-1.28	0.202	-0.51	0.11
15	log_budget	0.63	0.04	15.10	<0.001	0.55	0.71
16	log_vote_count	0.22	0.04	5.80	<0.001	0.15	0.30
17	log_popularity	0.27	0.04	6.20	<0.001	0.18	0.35
18	vote_average	0.22	0.02	11.25	<0.001	0.18	0.26
19	year_centered	-0.02	0.00	-6.00	<0.001	-0.03	-0.02
20	in_collection	0.43	0.03	14.54	<0.001	0.37	0.49

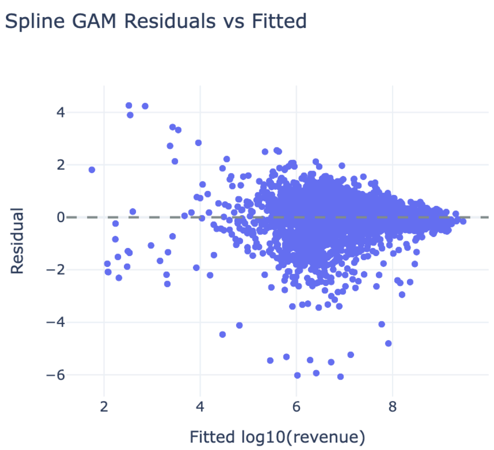


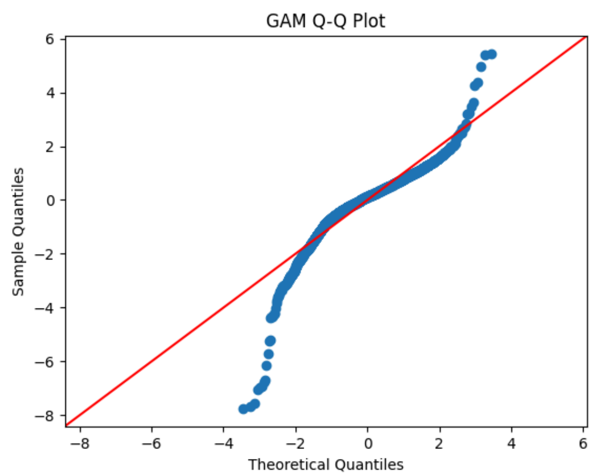


GAM



The partial effect plot for budget suggests a mostly increasing and only mildly nonlinear relationship with revenue. Although the GAM captures some curvature, particularly at the lowest and highest budget ranges, the effect remains broadly close to linear after log transformation.





We notice that the GAM exhibits very similar characteristics as the linear model: A spanning effect indicating signs of heteroscedasticity as well as a non normal residual.

Regression Performance Summary

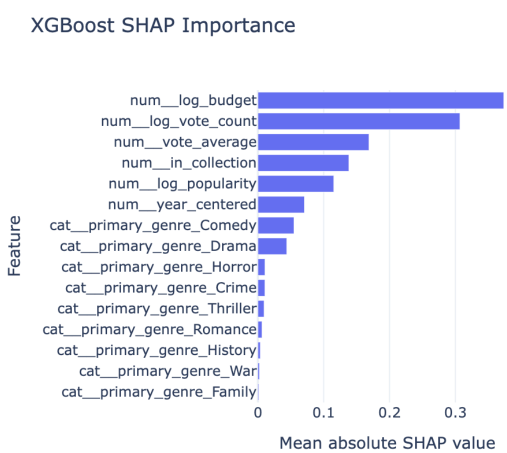
Model	Train R-squared	Adjusted R-squared	AIC	Test RMSE	Test MAE	Test R-squared
Linear regression	0.59	0.59	8,579.00	0.70	0.51	0.64
Spline GAM	0.61	0.61	8,466.80	0.69	0.50	0.65

Xgboost

XGBoost Best Parameters

colsample_bytree	learning_rate	max_depth	n_estimators	subsample	best_cv_rmse
0.800000	0.050000	3	200	0.800000	0.789000

	Model	Train RMSE	Train MAE	Train R-squared
0	Linear regression	0.80	0.55	0.59
1	Spline GAM	0.78	0.53	0.61
2	XGBoost	0.70	0.48	0.69



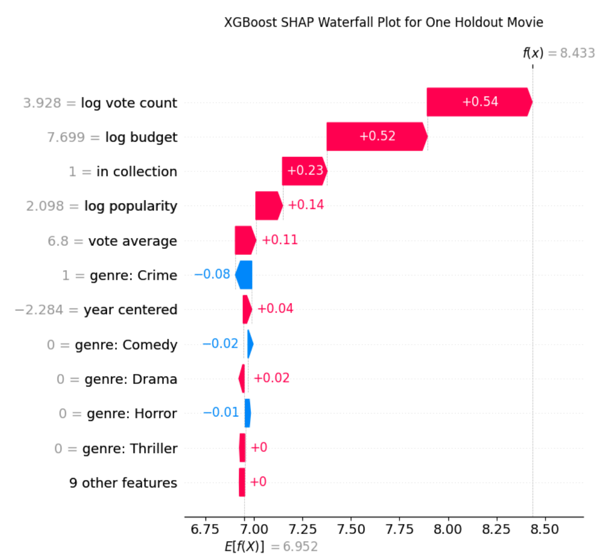


Figure X. SHAP waterfall plot for one holdout movie under the final XGBoost model. The plot starts at the baseline prediction, E[f(X)], which represents the average predicted log box office revenue across the sample, and shows how each feature shifts that prediction to the final movie-specific value, f(x). Red bars indicate features that increase the predicted revenue, while blue bars indicate features that decrease it. In this example, budget, vote count, and franchise membership contribute most strongly to a higher predicted revenue.

Summary and Conclusion

The linear regression model serves as the interpretability baseline. It shows that higher budget, higher vote count, and franchise membership are all positively associated with higher revenue, holding other variables constant. This model captures the main directional story clearly, but its additive structure assumes that the effect of each predictor is constant across the rest of the feature space.

The residual diagnostics indicate that the linear specification does not fully capture the structure of the data. In particular, the residuals are clearly non-normal and show some heteroskedasticity across the fitted range. This is not surprising given the heavy-tailed nature of movie revenue and supports comparing the linear model to more flexible alternatives such as GAM and XGBoost

As for GAM, it only modestly improved in sample fit and did not meaningfully outperform the linear model on held-out data. Its residual diagnostics also remained approximately identical to the linear version, with heavy tails and some heteroskedasticity still visible. The main takeaway is that although some nonlinear structure may be present, simple nonlinearity was not the primary limitation of the regression framework. Instead, the remaining error appears to be driven more by extreme box office outcomes and omitted predictors than by the linear functional form alone. Thus, these two models violate the assumptions

A comparison of the two models reinforces this point. The linear regression achieved a train R^2 of 0.59 and a test R^2 of 0.64, while the spline GAM increased these values only slightly to 0.61 and 0.65, respectively. Test error was also nearly identical, with the GAM improving RMSE from 0.70 to 0.69 and MAE from 0.51 to 0.50. The GAM also produced a lower AIC, which suggests somewhat better in sample fit, but the practical improvement was small. Taken together, these results suggest that the log transformed linear model already captured most of the usable signal in the dataset, and that the additional flexibility of the GAM provided only marginal gains. However, because regression assumptions are violated in both models, coefficient significance should be interpreted cautiously and used primarily as reference evidence for the direction and relative strength of associations rather than as definitive inferential results

XGBoost performed best overall and achieved the strongest test-set fit. The improvement was not dramatic, but it was consistent. The important point is not just that XGBoost had the highest R^2, but that it was able to represent relationships that the additive models could not. In practice, this means it can capture combinations of predictors more flexibly, such as cases where franchise membership matters more in some genres than others, or where the impact of budget depends on how visible a movie already is. In other words, XGBoost suggests that as opposed to revenue having an additive relationship between the features, it is shaped by combinations of scale, branding, and engagement.

The SHAP analysis supports this interpretation. Budget remained the dominant feature, but vote count, vote average, and collection membership also contributed substantially to predictions.

The genre level SHAP drilldown further showed that feature importance is not uniform across the dataset. Franchise branding appears to matter more for large-scale genres such as Science Fiction, Adventure, and Animation, while its role is much weaker for genres like Drama. This helps explain why the machine learning model outperformed the simpler regression models: it was better able to adapt its weighting of features across different parts of the movie market.

Taken together, the results show that commercial success is driven first by production scale, but not by production scale alone. Budget is the foundation, but audience engagement and franchise branding provide important additional signals. The consistency of the budget effect across all three models reinforces that point. At the same time, the stronger performance of XGBoost indicates that movie revenue is not governed by one uniform slope. Instead, the effect of one variable often depends on the context created by the others.

In conclusion, we see that budget is the dominant predictor of box office revenue, but audience engagement and franchise branding are essential secondary drivers, and their importance varies across genres. Linear regression captures the broad structure of this relationship, the GAM shows that some effects are meaningfully nonlinear, and XGBoost performs best because it can represent the combined and context-dependent effects of multiple predictors at once.

Limitations

This study has three main limitations. First, the TMDB dataset does not include several important business side predictors of box office performance, such as marketing expenditure, release scale, critic scores, or richer measures of public sentiment, so the models are necessarily built on an incomplete feature set. Second, the analysis relies on log transformed revenue and a restricted sample of films with reported positive budget and revenue values, which improves stability but also introduces selection bias toward more commercially documented movies and makes errors on the log scale harder to interpret in raw dollar terms. Third, each modeling approach carries its own assumptions: the linear regression imposes a simplified additive structure and only partially satisfies classical residual assumptions, the GAM allows nonlinearity but still assumes additivity unless interactions are specified manually, and XGBoost is more flexible but depends on careful tuning to avoid overfitting.